

Food as a Human–Planetary System

*An Integrative Review of Nutrition, Safety, Culture, Processing, Equity,
Technology, and Sustainability*

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Highlights

- Food is best understood as a coupled system linking biology, culture, livelihoods, markets, technology, public health, and ecosystems.
- The contemporary food challenge is a “multiple burden”: hunger and micronutrient deficiencies coexist with obesity, diet-related disease, unsafe food, unaffordable healthy diets, and environmental damage.
- Food processing is not inherently beneficial or harmful; outcomes depend on purpose, formulation, food structure, safety, degree of transformation, and the dietary pattern in which a product is consumed.
- No single intervention can transform food systems. Progress requires coordinated action across production, food safety, affordability, food environments, innovation, and loss-and-waste prevention.
- Culturally legitimate, locally adaptable solutions are more likely to be equitable, trusted, and durable than universal menus detached from place and livelihood.

Abstract

Food is simultaneously matter, meaning, infrastructure, livelihood, and ecological force. This integrative review develops a unified account of food that connects its biochemical role in human nutrition with food safety, processing, culture, affordability, technological change, and planetary sustainability. A targeted evidence synthesis was conducted using authoritative international assessments and peer-reviewed literature, prioritizing global reports and research published from 2015 to July 2026 while retaining selected earlier foundational works. The evidence reveals a food system that produces sufficient aggregate energy yet distributes health, risk, and environmental cost unevenly. In 2024, an estimated 673 million people faced hunger, about 2.3 billion experienced moderate or severe food insecurity, and 2.60 billion could not afford a healthy diet. Meanwhile, the World Health Organization’s 2026 estimates attribute 866 million illnesses and 1.52 million deaths annually to unsafe food. Food-system activity also contributes substantially to greenhouse-gas emissions, freshwater use, biodiversity pressure, and waste: 13.3% of food was estimated to be lost between post-harvest and pre-retail stages in 2023, while 1.05 billion tonnes were wasted at retail, food-service, and household levels in 2022. The review argues that the conventional separation of nutrition, agriculture, safety, environment, and culture creates fragmented policy and perverse trade-offs. It proposes an integrated framework organized around five levers: producing diverse and resilient foods; protecting safety through One Health systems; enabling affordability and healthy food environments; innovating responsibly in processing, fortification, fermentation, logistics, and data; and conserving resources by preventing loss and waste. The central conclusion is that “better food” cannot be defined by nutrients alone. It must be safe, nourishing, affordable, desirable, culturally legitimate, fairly produced, and ecologically viable.

Keywords: food systems; nutrition; food safety; dietary patterns; food processing; food security; sustainability; food culture; food technology; food loss and waste

Abbreviations

Table 1. Key abbreviations used in the manuscript

Abbreviation	Meaning
DALY	Disability-adjusted life year
FAO	Food and Agriculture Organization of the United Nations
FBDG	Food-based dietary guideline
FIES	Food Insecurity Experience Scale
GHG	Greenhouse gas
HACCP	Hazard Analysis and Critical Control Point
NCD	Noncommunicable disease
NOVA	Food-processing classification system
SDG	Sustainable Development Goal
SOFI	State of Food Security and Nutrition in the World
UNEP	United Nations Environment Programme
UPF	Ultra-processed food
WHO	World Health Organization

1. Introduction

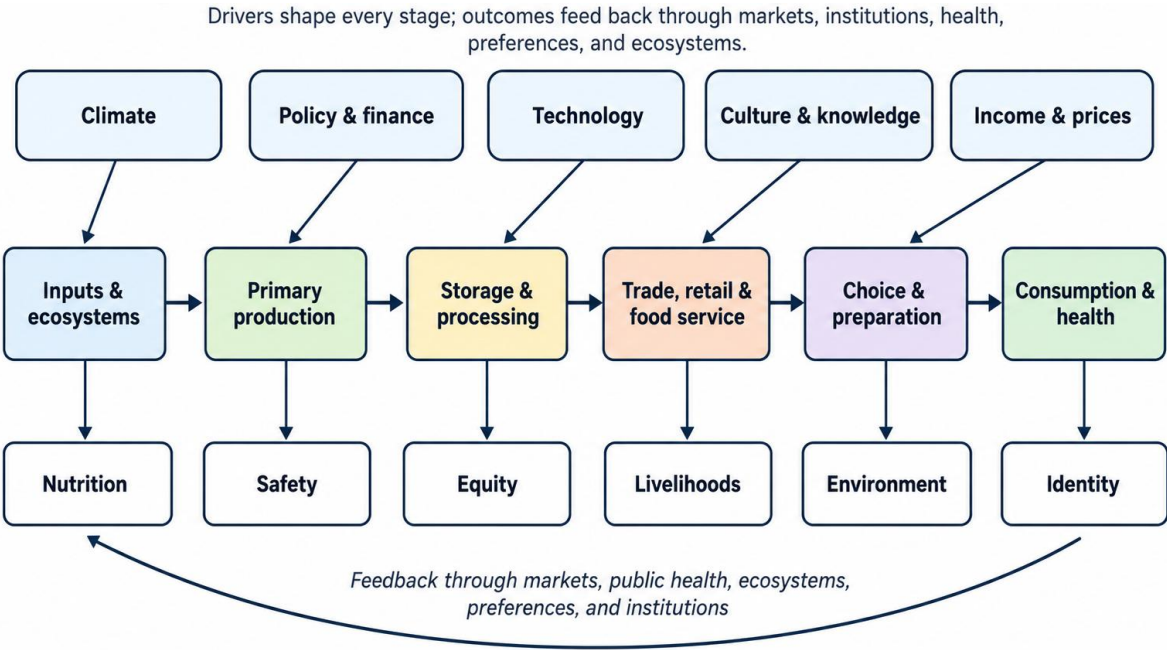
Food is often reduced to one of three things: a collection of nutrients, a market commodity, or a personal choice. Each description is useful, but none is sufficient. Food begins in soils, waters, seeds, animals, microorganisms, and human labour; it moves through storage, processing, logistics, markets, kitchens, institutions, and bodies; and it returns to the environment as emissions, wastewater, packaging, and organic residues. At every stage, food is shaped by power, knowledge, culture, and technology. A bowl of rice, a fermented milk, a pulse stew, a packaged snack, and a cultivated-protein product therefore differ not only in composition but also in social meaning, risk profile, resource demand, accessibility, and place within a dietary pattern.

The global evidence demonstrates why an integrated perspective is necessary. The 2025 edition of The State of Food Security and Nutrition in the World estimated that 673 million people faced hunger in 2024, approximately 2.3 billion experienced moderate or

severe food insecurity, and 2.60 billion were unable to afford a healthy diet (FAO et al., 2025). At the same time, unhealthy dietary patterns contribute materially to noncommunicable disease, and adult obesity continues to rise in many regions. Food must also be safe: the WHO’s updated burden estimates indicate that contaminated food caused approximately 866 million illnesses, 1.52 million deaths, and 57.1 million healthy life-years lost in 2021 (WHO, 2026a). These burdens overlap rather than occur in isolation. A low-income household may face insufficient food quantity, micronutrient inadequacy, exposure to unsafe food, and high consumption of cheap energy-dense products within the same week.

Food is also a major driver of environmental change. A widely used global inventory estimated that food systems accounted for 34% of anthropogenic greenhouse-gas emissions in 2015, including agriculture, land-use change, processing, transport, retail, consumption, and waste (Crippa et al., 2021). Agriculture is the largest user of freshwater withdrawals globally, while food production is a major pressure on land, nutrient cycles, and biodiversity. Loss and waste magnify these impacts because all resources embodied in uneaten food have been used without delivering nutrition. The food question is thus not simply how to produce more, but how to produce, distribute, prepare, consume, and value food differently.

This review has four objectives. First, it clarifies the multidimensional nature of food. Second, it synthesizes evidence across nutrition, safety, culture, processing, security, technology, and sustainability. Third, it identifies recurrent tensions—for example, affordability versus nutrient density, shelf life versus formulation quality, productivity versus ecological integrity, and global standards versus cultural diversity. Fourth, it proposes a practical framework for coordinated food-system transformation.



Original synthesis by SAMUELSON G.

Figure 1. Food as a coupled human–planetary system. The diagram is an original synthesis by the author. It depicts the food chain as a flow embedded in upstream drivers and downstream outcomes, with feedback through markets, health, institutions, preferences, and ecosystems.

2. Review Scope and Methods

This article is an integrative review and conceptual synthesis rather than a systematic review or meta-analysis. The approach was selected because the topic “food” spans multiple evidence traditions—including nutritional epidemiology, clinical trials, food science, microbiology, anthropology, economics, agricultural science, environmental assessment, and public policy—that cannot be adequately represented by a single intervention, exposure, or outcome framework.

A targeted search and source-verification process was conducted in July 2026. Priority was given to: (1) global and normative assessments from FAO, WHO, UNICEF, WFP, UNEP, IPCC, the Codex Alimentarius Commission, and the High Level Panel of Experts on Food Security and Nutrition; (2) peer-reviewed reviews, commissions, randomized trials, cohort syntheses, and global modelling studies; and (3) seminal works needed to explain food culture, cooking, and food chemistry. Literature from 2015–2026 was emphasized, with earlier foundational works retained when conceptually indispensable. Searches used combinations of terms relating to healthy diets, dietary patterns, ultra-processed foods, foodborne disease, food security, affordability, food-system emissions, water use, food loss and waste, fortification, fermentation, and emerging food technology.

Sources were included when they had clear methods, global or cross-context relevance, direct bearing on the manuscript’s conceptual model, or authoritative status in international guidance. The review intentionally distinguishes between measured historical estimates, modelled scenarios, and normative recommendations. Because the evidence base is large and heterogeneous, the synthesis should not be interpreted as exhaustive. No new human, animal, or laboratory data were collected, and no pooled effect estimates were calculated.

3. What Is Food?

3.1 Biological substrate and nutritional function

At the biological level, food supplies energy, indispensable amino acids and fatty acids, vitamins, minerals, water, fibre, and a vast range of bioactive compounds. These materials support growth, maintenance, immune function, reproduction, cognition, movement, and repair. Yet nutrient composition alone does not predict the full physiological effect of a food. Structure, particle size, cooking, ripeness, food matrix,

fibre integrity, fermentation, combinations with other foods, and eating rate can alter digestibility, glycaemic response, satiety, and bioavailability. A whole grain and a refined flour may contain chemically related carbohydrates but produce different physiological responses because the physical matrix and accompanying fibre differ.

The FAO–WHO joint statement defines healthy diets through four universal principles: adequacy, balance, moderation, and diversity (FAO & WHO, 2024). These principles are deliberately broader than numerical nutrient targets. Adequacy protects against deficiency; balance aligns energy and nutrient sources with need; moderation constrains components whose excess increases risk; and diversity raises the likelihood of meeting micronutrient requirements while supporting dietary resilience. WHO guidance further emphasizes whole grains, vegetables, fruits, pulses, nuts, and suitable protein sources, while limiting free sugars, sodium, unhealthy fats, and industrial trans fats (WHO, 2026b).

3.2 Food as transformed matter

Almost all human diets depend on processing. Washing, cutting, grinding, soaking, fermenting, cooking, drying, chilling, freezing, canning, and packaging can improve safety, digestibility, flavour, shelf life, convenience, and nutrient access. Heat gelatinizes starch, denatures proteins, softens plant cell walls, destroys many pathogens, and creates aromas through reactions such as Maillard chemistry. Fermentation can preserve food, modify flavour and texture, reduce some antinutritional factors, and introduce live microorganisms or metabolites with possible health effects (Marco et al., 2017; Wastyk et al., 2021). Fortification and biofortification can address population-level micronutrient deficiencies when implemented with appropriate monitoring (Allen et al., 2006; Bouis & Saltzman, 2017).

Processing becomes problematic when it is used primarily to engineer products that are energy-dense, rapidly consumed, aggressively marketed, and nutritionally imbalanced. The NOVA framework calls many industrial formulations “ultra-processed” when they contain combinations of refined substances, additives, and processes not normally used in domestic cooking (Monteiro et al., 2019). Observational syntheses associate higher UPF exposure with multiple adverse outcomes, although certainty varies by outcome and residual confounding remains important (Lane et al., 2024). A tightly controlled inpatient trial nevertheless demonstrated a causal effect on short-term intake: participants consumed more energy and gained weight on an ultra-processed diet than on a minimally processed diet despite meals being matched for presented calories, macronutrients, sugar, sodium, and fibre (Hall et al., 2019). The correct inference is not that every packaged or industrially processed food is harmful, but that formulation, structure, palatability, eating rate, marketing, and substitution within the total diet matter.

3.3 Food as culture, identity, and social relationship

Food is a language of belonging. Cuisines encode histories of migration, ecology, trade, religion, class, gender, colonization, celebration, scarcity, and adaptation. Staples often anchor identity; spices and techniques distinguish regions; meal timing structures social life; and rules about purity, hospitality, fasting, and commensality define moral communities. Mintz’s study of sugar showed how a familiar food can connect plantation labour, empire, industrialization, and changing patterns of taste (Mintz, 1985). Food knowledge is therefore not merely a collection of recipes. It is a living archive of environmental observation, preservation methods, sensory standards, and social obligations.

Cultural legitimacy is a practical condition for nutrition policy. A technically optimal diet that ignores taste, ritual, status, labour time, local availability, gendered workloads, and culinary skill is unlikely to be adopted or sustained. Conversely, traditional diets should not be romanticized as automatically healthy or sustainable; traditions also change, can reproduce exclusion, and may contain high-risk practices. The policy objective is not to freeze cuisines but to support culturally grounded improvement—for example, increasing the proportion of vegetables and pulses in familiar dishes, reformulating commonly eaten foods, preserving beneficial fermentation practices, and protecting indigenous and local food knowledge while ensuring safety.

Table 2. Six analytical dimensions of food

Dimension	Central question	Typical evidence	Illustrative outcome	Frequent blind spot
Biological	What does the food provide to and do within the body?	Biochemistry, physiology, clinical trials, epidemiology	Growth, satiety, deficiency prevention, disease risk	Treating foods as isolated nutrients
Safety	What hazards can occur from production to consumption?	Microbiology, toxicology, surveillance, risk assessment	Illness, mortality, outbreak prevention	Separating human, animal, plant, and environmental health
Cultural	What does the food mean, and how is it prepared and shared?	Anthropology, history, sensory studies	Identity, trust, adoption, social cohesion	Assuming preferences are fixed or purely individual
Economic	Who can produce,	Price data,	Affordability,	Equating

Dimension	Central question	Typical evidence	Illustrative outcome	Frequent blind spot
	buy, and benefit from the food?	household surveys, value-chain analysis	wages, livelihoods, resilience	market availability with access
Technological	How is food transformed, preserved, distributed, and verified?	Food science, engineering, logistics, data science	Shelf life, convenience, quality, traceability	Treating novelty as inherently beneficial
Ecological	What resources, emissions, pollutants, and biodiversity effects are embodied?	Life-cycle assessment, inventories, Earth-system modelling	GHG emissions, land and water use, waste	Ignoring local context and trade-offs

4. Nutrition, Dietary Patterns, and Health

4.1 From nutrient sufficiency to dietary quality

Nutrition science historically focused on identifying deficiency diseases and the nutrients required to prevent them. That work enabled major advances, including iodized salt, folic-acid fortification, vitamin A programmes, and treatment of severe acute malnutrition. The contemporary challenge is broader. Diets must prevent deficiency without promoting chronic excess, and they must do so across the life course. The same population can experience child stunting, anaemia, adult obesity, diabetes, and cardiovascular disease—a “multiple burden” created by poverty, unequal food environments, infection, inadequate care, sedentary work, and the widespread availability of cheap, energy-dense foods.

Dietary patterns provide a more useful unit of analysis than single nutrients because foods are consumed in combinations and substitutions. Evidence consistently supports patterns rich in vegetables, fruits, whole grains, legumes, nuts, and unsaturated fats, with appropriate protein sources and limited sodium, free sugars, processed meats, and industrial trans fats. In the PREDIMED trial, a Mediterranean dietary pattern supplemented with extra-virgin olive oil or nuts reduced major cardiovascular events among high-risk adults compared with a lower-fat control diet (Estruch et al., 2018). A large systematic review and meta-analysis found that higher intakes of dietary fibre and whole grains were associated with lower incidence and mortality from several noncommunicable diseases (Reynolds et al., 2019). Global comparative risk assessment

has likewise identified low intake of whole grains and fruits and high sodium intake among major diet-related risks (Afshin et al., 2019).

4.2 Life-course and context specificity

Healthy diets are not identical for all people. Requirements and acceptable trade-offs vary with age, pregnancy, lactation, disease, disability, physical activity, climate, and food access. Infants require breastfeeding support and safe, nutrient-dense complementary foods; adolescents need adequate energy and nutrients during rapid growth; older adults may require greater attention to protein quality, food texture, hydration, and social isolation. In some low-resource contexts, modest increases in nutrient-dense animal-source foods can improve the diets of children and pregnant or lactating women. In affluent contexts with high red and processed meat intake, shifting toward legumes, nuts, fish where appropriate, and other plant-forward sources can reduce health and environmental burdens. Context therefore determines the direction and magnitude of recommended change.

4.3 Food environments and the limits of individual choice

Food choice is constrained by price, time, transport, cooking facilities, refrigeration, safety, marketing, school and workplace offerings, retail placement, social norms, and stress. The least nutritious option may also be the most convenient, shelf-stable, predictable, and socially promoted. Healthy-diet policy must therefore modify environments rather than rely only on education. Evidence-informed tools include healthy public procurement, restrictions on marketing to children, clear front-of-pack labelling, reformulation targets, elimination of industrial trans fats, sodium reduction, taxes on sugar-sweetened beverages, subsidies or incentives for nutrient-dense foods, breastfeeding protection, and primary-care nutrition support. The distributional effect of each policy should be assessed: a tax can improve health but become regressive if affordable alternatives and income support are absent.

Table 3. Nutritional roles, food sources, and principal concerns

Component	Core functions	Food sources	Deficiency or low-intake concerns	Excess or imbalance concerns
Carbohydrate and fibre	Energy; fibre supports bowel function, microbiota, satiety, and metabolic health	Whole grains, roots and tubers, fruits, vegetables, pulses	Low energy intake; low fibre; inadequate whole grains	Excess refined starches and free sugars; displacement of nutrient-dense foods

Component	Core functions	Food sources	Deficiency or low-intake concerns	Excess or imbalance concerns
Protein	Tissue synthesis, enzymes, hormones, immunity	Pulses, dairy, eggs, fish, meat, nuts, seeds, grains	Growth failure; loss of lean mass; poor diet quality	Unbalanced sources; high processed-meat intake; excessive intake in susceptible individuals
Fats	Energy, cell membranes, essential fatty acids, absorption of fat-soluble vitamins	Oils, nuts, seeds, fish, dairy, meat, avocado	Essential-fatty-acid deficiency; inadequate energy	Excess saturated fat; industrial trans fat; energy excess
Vitamins and minerals	Metabolism, oxygen transport, bone health, thyroid function, immunity, development	Diverse plant and animal foods; fortified foods	Anaemia; iodine deficiency; vitamin A, D, B12, folate, zinc and other deficiencies	Toxicity from excessive supplements; sodium excess
Water and electrolytes	Hydration, temperature regulation, circulation, cellular function	Water, beverages, fruits, vegetables, prepared foods	Dehydration; unsafe drinking water	Excess sodium; sugar-sweetened beverage intake
Bioactive compounds and food matrix	Modulate sensory, microbial, metabolic and physiological responses	Herbs, spices, fruits, vegetables, fermented foods, intact grains	Loss of diversity and matrix through refining	Claims may exceed evidence; interactions and dose remain uncertain

5. Food Safety: From Hazard Control to One Health

Food cannot be healthy if it is unsafe. Hazards may be biological, chemical, physical, or allergenic, and they can enter at any stage—from contaminated irrigation water and animal feed to processing equipment, transport, retail handling, or domestic kitchens.

WHO's 2026 global estimates attribute 866 million illnesses and 1.52 million deaths to contaminated food in 2021; children under five bear a disproportionate share of the burden (WHO, 2026a). These figures also underestimate the social experience of foodborne illness because surveillance and attribution remain incomplete in many countries.

The most effective systems prevent hazards rather than relying solely on end-product testing. Good agricultural, veterinary, manufacturing, and hygienic practices establish baseline controls; HACCP identifies critical points where hazards can be prevented, eliminated, or reduced; risk-based inspection focuses limited capacity on the greatest threats; and traceability enables rapid recall. The Codex General Principles of Food Hygiene provide the international foundation for these practices (Codex Alimentarius Commission, 2022). At household and food-service levels, safe water, hand hygiene, separation of raw and cooked foods, adequate cooking, temperature control, and safe sourcing remain essential.

A One Health approach recognizes that food safety cannot be separated from animal health, plant health, environmental contamination, antimicrobial resistance, climate, and trade. Rising temperatures can alter pathogen growth and toxin formation; floods can contaminate fields and water systems; drought may concentrate some contaminants; and longer, more complex supply chains can spread local failures across borders. Whole-genome sequencing, interoperable surveillance, and international notification improve detection, but they must be matched by laboratory capacity, transparent governance, and support for small producers to comply without being excluded from markets (FAO, UNEP, WHO, & WOAHA, 2022).

Table 4. Major food-safety hazard classes and control strategies

Hazard class	Examples	Common routes	Potential effects	Priority controls
Bacterial	Campylobacter, Salmonella, pathogenic E. coli, Listeria	Raw or undercooked foods, cross-contamination, poor temperature control	Gastroenteritis, invasive disease, miscarriage, death	Hygiene, pasteurization/cooking, cold chain, environmental monitoring, HACCP
Viral	Norovirus, hepatitis A and E	Infected handlers, contaminated water, shellfish, produce	Acute gastroenteritis, hepatitis	Hand hygiene, safe water, exclusion of ill handlers, sanitation
Parasitic	Toxoplasma,	Undercooked	Acute and	Veterinary control,

Hazard class	Examples	Common routes	Potential effects	Priority controls
	Taenia, Cryptosporidium, foodborne trematodes	meat/fish, contaminated water or produce	chronic infection, neurological or organ damage	water safety, cooking/freezing where effective
Chemical	Mycotoxins, heavy metals, pesticide residues, cleaning chemicals	Contaminated crops, water, equipment, misuse of chemicals	Acute poisoning, cancer, developmental and organ effects	Prevention at source, standards, monitoring, storage and moisture control
Allergenic	Peanut, milk, egg, wheat, soy, tree nuts, fish, shellfish and others	Undeclared ingredients, cross-contact, labelling failure	Anaphylaxis and other immune reactions	Supplier control, segregation, validated cleaning, accurate labelling
Physical	Glass, metal, stones, hard plastic, bone fragments	Equipment damage, raw materials, poor handling	Choking, cuts, dental injury	Equipment maintenance, sieves, magnets, detection systems, visual control

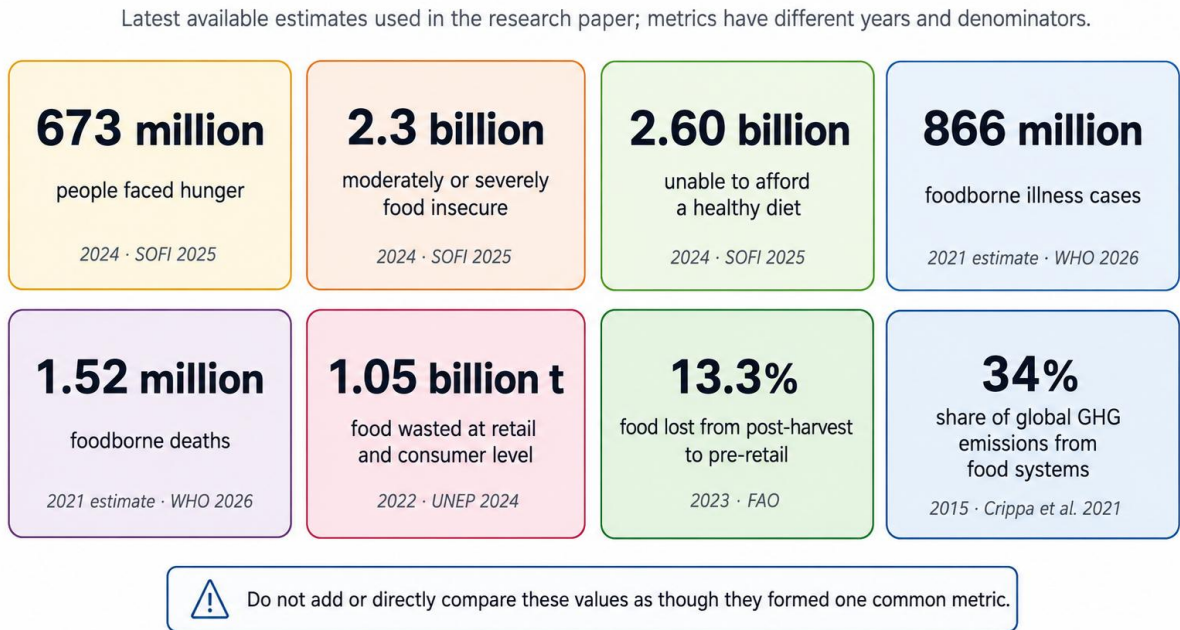
6. Food Security, Affordability, and Equity

Food security exists when people have reliable physical, social, and economic access to sufficient, safe, and nutritious food that meets dietary needs and preferences for an active and healthy life. Contemporary frameworks also emphasize agency—the ability of people to shape food systems—and sustainability over time (HLPE, 2020). These dimensions expose why increasing aggregate supply is necessary but insufficient. Food may be present in a country yet inaccessible to households because of price, income, discrimination, conflict, displacement, transport barriers, or lack of cooking facilities.

The latest global estimates show modest improvement in average hunger alongside severe regional divergence. SOFI 2025 estimated that between 638 and 720 million people faced hunger in 2024, with a point estimate of 673 million. About 2.3 billion experienced moderate or severe food insecurity, and 2.60 billion—31.9% of the global population—could not afford a healthy diet (FAO et al., 2025). The global decline in unaffordability concealed deterioration in Africa and low-income countries. High food-price inflation is especially damaging to poor households because food absorbs a large share of expenditure and because nutrient-dense foods such as fruits, vegetables, and animal-source foods can be relatively expensive per unit of energy.

Affordability is not only a consumer problem. Farmers and food workers may produce food while remaining food insecure because of low or volatile income, insecure land tenure, high input costs, weak bargaining power, and climate exposure. Women frequently face unequal access to land, finance, extension, and decision-making while performing substantial unpaid food and care work. Indigenous peoples and pastoral, fishing, forest, and smallholder communities can lose access to territories and foods through conservation, infrastructure, or market policies designed without participation. A rights-based food policy therefore asks both whether people can eat well and whether the system distributes voice, risk, labour, and value fairly.

Price shocks require layered responses. In the short term, targeted cash transfers, vouchers, school meals, maternal and child nutrition services, and temporary relief can protect consumption. In the medium term, storage, transport, cold chains, market information, competition policy, and diversified domestic and regional supply can reduce volatility. Long-term resilience requires investment in research, climate adaptation, rural infrastructure, public health, and social protection. Broad price controls and export restrictions may provide immediate political relief but can distort incentives, reduce supply, and export instability to neighbouring countries when poorly designed (FAO et al., 2025).



Sources: FAO et al. (2025); WHO (2026); UNEP (2024); FAO (2025); Crippa et al. (2021).

Figure 2. Selected global food-system burdens using the latest available authoritative estimates. The values refer to different years, populations, denominators, and methods; the figure is a dashboard rather than an additive comparison. Sources: FAO et al. (2025), WHO (2026a), UNEP (2024), FAO (2025), and Crippa et al. (2021).

7. Food and the Biosphere

7.1 Climate, land, water, and biodiversity

Food systems are both vulnerable to environmental change and important drivers of it. Climate shocks affect yields, fisheries, livestock, labour productivity, storage, transport, prices, and food safety. At the same time, agriculture and land-use change release carbon dioxide, methane, and nitrous oxide; energy is used in processing, refrigeration, transport, retail, and cooking; and waste emits methane when landfilled. Crippa et al. (2021) estimated global food-system emissions at 18 Gt CO₂-equivalent in 2015, 34% of total anthropogenic emissions. The estimate includes pre- and post-production processes that are sometimes omitted from agricultural inventories.

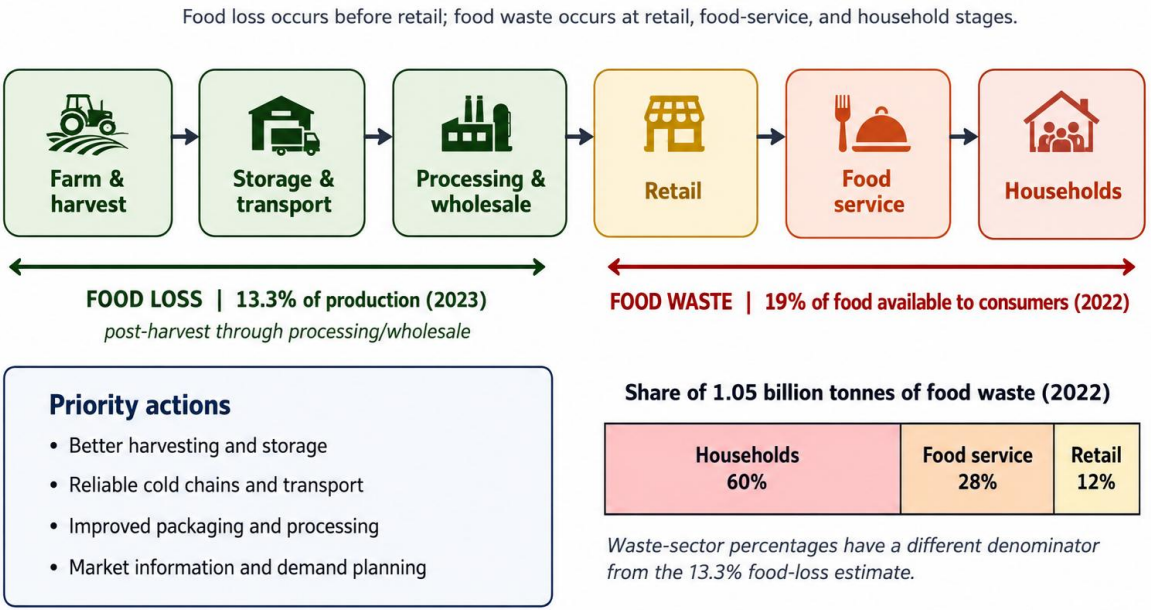
Environmental intensity varies widely among products and production systems. Life-cycle assessment generally finds that ruminant meat has high median greenhouse-gas and land footprints, although performance varies by system; many plant proteins have lower impacts per unit of protein (Poore & Nemecek, 2018). Yet simple product rankings can obscure nutrient density, livelihoods, grazing ecology, co-products, local water scarcity, and cultural roles. The most defensible strategy is portfolio-based: reduce the highest avoidable impacts, improve production practices, shift consumption where excess is greatest, protect nutrient access where deficiency persists, and prevent rebound effects.

Food-system pathways consistent with climate and biodiversity goals combine dietary change, yield improvement, lower input losses, protection and restoration of ecosystems, reduced food loss and waste, and changes in technology and management (Springmann et al., 2018; Clark et al., 2020; Rockström et al., 2025). Diet shifts should not be framed as a single global menu. Plant-forward patterns can be expressed through diverse cuisines and should be adapted to local nutritional needs, agricultural ecologies, and incomes. The updated EAT–Lancet Commission emphasizes healthy, sustainable, and just food systems, adding equity and rights to earlier health-and-environment framing (Rockström et al., 2025).

7.2 Food loss, waste, and circularity

Food loss occurs before retail, often because of harvesting damage, inadequate drying, pests, poor storage, weak cold chains, processing inefficiency, or market rejection. Food waste occurs in retail, food service, and households through over-purchasing, oversized portions, confusing date labels, strict cosmetic standards, poor inventory management, and inadequate storage or meal planning. FAO estimated that 13.3% of food was lost from post-harvest through pre-retail stages in 2023, with the highest proportional losses in fruits and vegetables. UNEP estimated that 1.05 billion tonnes—19% of food available to consumers—were wasted in 2022; households accounted for 60%, food service for 28%, and retail for 12% (UNEP, 2024; FAO, 2025).

The prevention hierarchy matters. The first priority is to avoid surplus and spoilage; the second is to redistribute safe edible food; the third is to redirect appropriate materials to animal feed or industrial uses; and only then to recycle nutrients through composting or anaerobic digestion. Disposal is the least preferred option. Technologies such as hermetic storage, evaporative cooling, refrigerated transport, demand forecasting, dynamic pricing, smaller portions, improved date labelling, and waste measurement can help. However, the causes are institutional as well as technical: contracts, procurement specifications, liability rules, promotions, labour practices, and consumer expectations determine what becomes waste.



Sources: FAO (2025) and UNEP (2024).

Figure 3. Food loss and waste across the supply chain. Food-loss and food-waste percentages have different denominators and should not be summed mechanically. Waste-sector shares refer to the UNEP estimate of 1.05 billion tonnes in 2022. Sources: FAO (2025) and UNEP (2024).

8. Food Processing and Emerging Technology

Technology has always shaped food systems. Fire, grinding stones, fermentation vessels, irrigation, milling, refrigeration, canning, pasteurization, synthetic fertilizer, breeding, packaging, and digital logistics each altered what people could eat, where, when, and at what cost. The relevant question is therefore not whether food should be technological, but which technologies create public value, for whom, under what safeguards, and with which unintended consequences.

8.1 Established high-value technologies

Many of the highest-impact technologies are unglamorous: safe water, sanitation, drying, storage, refrigeration, pasteurization, heat treatment, packaging that prevents contamination, quality testing, and reliable transport. Fortification can deliver iodine, iron, folic acid, vitamin A, and other micronutrients through widely consumed staples or condiments, provided that nutrient forms, dosing, quality assurance, coverage, and surveillance are appropriate. Biofortification uses plant breeding or agronomy to increase nutrient concentrations in staple crops and can complement dietary diversification (Allen et al., 2006; Bouis & Saltzman, 2017). Automation and sensors can improve precision, labour safety, resource efficiency, and traceability, but adoption can deepen inequality when capital, connectivity, skills, and data rights are uneven (FAO, 2022).

8.2 Fermentation, alternative proteins, and cellular production

Fermentation ranges from traditional microbial ecosystems to controlled production of enzymes, flavours, fats, and proteins. It can enhance preservation and sensory diversity and may provide health benefits through live microbes or fermentation-derived metabolites, although effects are product- and population-specific. Precision fermentation and biomass fermentation may produce ingredients with lower land requirements than conventional livestock in some scenarios, but impacts depend on feedstocks, energy, purification, scale, and displacement. Plant-based meat analogues can reduce some environmental pressures relative to high-impact meat while varying greatly in sodium, fat, protein quality, additives, cost, and consumer acceptance.

Cultivated meat and other cell-based foods seek to produce animal tissue from cells rather than whole animals. Potential benefits include reduced slaughter, controlled production, and lower land use, while uncertainties remain around energy demand, growth media, scale, contamination control, cost, regulation, and consumer trust (Stephens et al., 2018). These technologies should be assessed against realistic comparators and complete life cycles, not idealized future systems or the worst current production. They are potential components of transformation, not substitutes for improvements in agriculture, diets, public health, and waste prevention.

8.3 Data, artificial intelligence, and governance

Digital tools can forecast demand, detect crop stress, optimize irrigation and routes, monitor cold chains, identify contamination patterns, support recall, automate waste measurement, and personalize dietary advice. Artificial intelligence can help discover formulations and predict sensory or functional properties, but food data are fragmented, proprietary, culturally biased, and uneven in quality. Algorithms may optimize the

metric they are given while ignoring unmeasured values such as culinary identity, labour, affordability, biodiversity, or long-term health. Responsible use requires validation in the intended setting, cybersecurity, privacy protection, transparent objectives, human oversight, and access for small producers and public agencies—not only dominant firms.

Table 5. Assessment of selected food technologies

Technology	Primary value proposition	Potential benefits	Principal risks or uncertainties	Responsible-use conditions
Cold chains and improved storage	Preserve quality and safety	Lower loss; longer market access; safer perishables	Energy demand, refrigerants, capital and maintenance	Efficient equipment, low-impact refrigerants, reliable power, inclusive finance
Fortification and biofortification	Increase micronutrient intake	Population reach; relatively low marginal cost	Over- or under-dosing; uneven coverage; reduced emphasis on diversity	Standards, quality assurance, biomarker surveillance, dietary complementarity
Traditional and controlled fermentation	Preserve and transform foods	Flavour, shelf life, possible microbiome and nutritional effects	Contamination, variable composition, overstated health claims	Hygiene, starter control where needed, product-specific evidence, cultural stewardship
Plant-based analogues	Replace selected animal products	Potentially lower land and GHG burdens; familiar formats	Nutritional variability, sodium, cost, processing, consumer trust	Transparent labelling, nutritional benchmarks, life-cycle assessment, affordability
Precision fermentation	Produce specific proteins, fats or ingredients	Functional design; potentially lower land use	Energy, feedstocks, purification, intellectual property, scale	Renewable energy, open safety data, realistic comparative assessment
Cultivated cell-based	Produce animal tissue	Animal-welfare potential;	High cost, energy use, media, scale,	Independent assessment, robust

Technology	Primary value proposition	Potential benefits	Principal risks or uncertainties	Responsible-use conditions
foods	without raising whole animals	controlled composition and production	safety and acceptance	regulation, transparent claims, staged deployment
AI and digital traceability	Improve prediction, control, and information	Efficiency, outbreak detection, waste reduction, personalized support	Bias, privacy, cybersecurity, market concentration, false precision	Human oversight, interoperability, public-interest governance, local validation

9. Governance, Ethics, and the Political Economy of Food

Food systems are governed by rules about land, water, labour, trade, subsidies, intellectual property, competition, safety, labelling, marketing, procurement, welfare, and environmental protection. These rules often pursue conflicting objectives. Agricultural policy may reward commodity volume while health policy asks people to eat more diverse diets. Trade policy may lower average prices while increasing exposure to shocks. Safety rules may protect consumers yet impose fixed compliance costs that exclude small vendors. Environmental standards may reduce emissions but shift production or raise prices if transition support is absent.

Policy coherence begins by making objectives and trade-offs explicit. Subsidies and public research can be redirected toward fruits, vegetables, pulses, nuts, resilient staples, sustainable livestock and fisheries, soil health, water efficiency, and diversified local value chains. Public procurement—especially school meals, hospitals, and social programmes—can create stable demand for healthy foods while shaping markets. Competition and transparency policy can address excessive concentration and unfair contracting. Labour standards are food policy because safe, fairly paid workers are essential to resilient production and processing.

Commercial influence deserves particular attention. Food firms contribute investment, innovation, logistics, and employment, but their fiduciary incentives may favour products with high margins, long shelf lives, and repeat purchase rather than population health. Conflict-of-interest safeguards, lobbying transparency, independent evaluation, restrictions on marketing to children, and enforceable rather than voluntary standards are therefore important. At the same time, regulation must distinguish among firms and products and avoid creating barriers that only large companies can absorb.

Ethical food governance also requires participation. Farmers, fishers, food workers, women's groups, indigenous peoples, consumers, public-health professionals, small businesses, and local governments hold different knowledge and bear different risks. Consultation after policy design is not sufficient; affected groups need meaningful roles in problem definition, option assessment, implementation, and evaluation. Data should be disaggregated by income, gender, age, geography, disability, and livelihood so that average improvements do not conceal widening inequality.

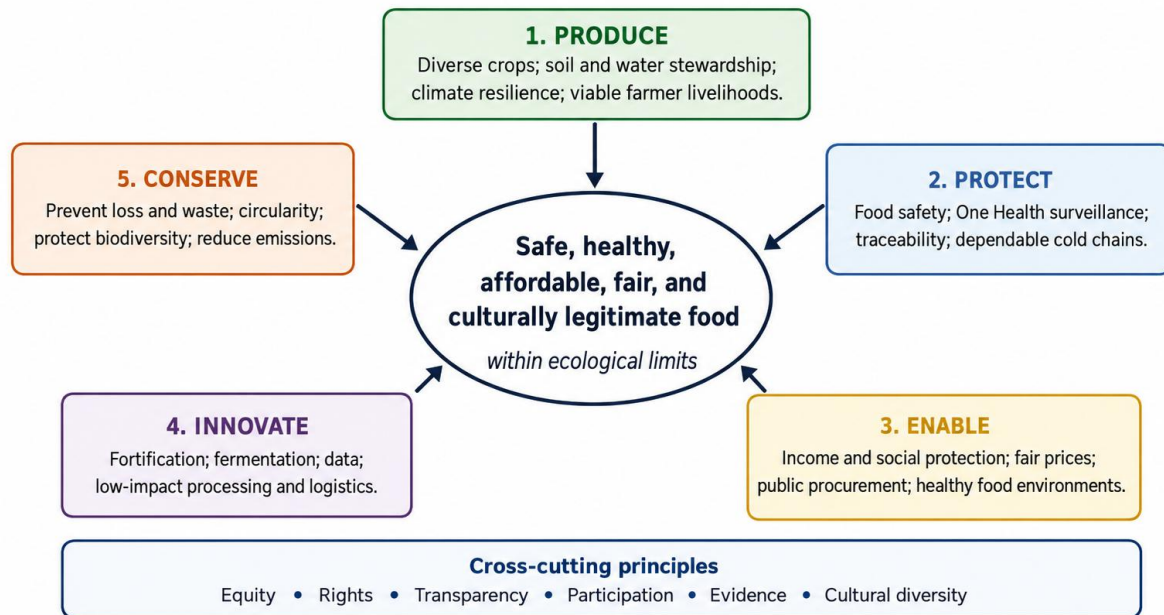
10. An Integrated Framework for Transformation

The evidence supports a five-lever framework. The levers are interdependent: action in one area can amplify or undermine another. For example, increasing fruit and vegetable production has limited effect if post-harvest losses remain high, retail prices are unaffordable, food-safety controls are weak, or consumers lack time and facilities to prepare the foods. Likewise, reformulation may improve nutrient composition but leave marketing, portion size, and displacement of minimally processed foods unchanged.

1. Produce diverse and resilient foods. Protect soil, water, pollinators, genetic diversity, and farmer livelihoods; adapt production to climate risk; and align research and incentives with dietary needs rather than commodity volume alone.
2. Protect safety from farm to table. Build One Health surveillance, laboratories, hygienic infrastructure, traceability, risk-based inspection, and practical training for businesses and households.
3. Enable affordability and healthy choice. Strengthen incomes and social protection, shape school and workplace food, regulate harmful marketing, improve labels, and ensure that healthy options are convenient and desirable.
4. Innovate for public value. Use processing, fortification, fermentation, logistics, biotechnology, automation, and data to solve defined problems, with independent assessment, transparency, and equitable access.
5. Conserve resources and prevent loss and waste. Measure losses, redesign contracts and portions, improve storage and forecasting, redistribute safe surplus, recover nutrients, and internalize environmental costs without reducing food access.

Three tests should be applied to every intervention. First is the substitution test: what will the intervention replace in practice? A plant-based product that displaces legumes may have a different effect from one that displaces high-impact processed meat. Second is the distribution test: who gains, who pays, and who bears transition risk? Third is the systems test: does the intervention improve multiple outcomes or merely shift harm across health, environment, geography, or time?

Five mutually dependent levers organized around a shared public goal



Original synthesis by SAMUELSON G.

Figure 4. Integrated framework for food-system transformation. The central goal combines safety, health, affordability, cultural legitimacy, fairness, and ecological viability. The surrounding levers are mutually dependent and should be evaluated using substitution, distribution, and systems tests. Original synthesis by the author.

Table 6. Policy and practice intervention matrix

Level	High-priority actions	Primary outcomes	Equity safeguards	Example indicators
Household and community	Culinary education; safe handling; breastfeeding support; meal planning; community gardens and food-sharing systems	Diet quality, safety, reduced waste, social connection	Time, fuel, water, disability and gender constraints addressed	Diet diversity; household waste; foodborne incidents; participation
Institutional	Healthy procurement; water and sanitation; portion and menu redesign; waste measurement; nutrition standards	Healthy food environments; stable demand; lower waste	Free or subsidized access; culturally appropriate menus; worker participation	Meal quality; uptake; plate waste; procurement share from priority foods
Municipal	Markets, cold	Access, local	Support for	Travel time to

Level	High-priority actions	Primary outcomes	Equity safeguards	Example indicators
and regional	chains, transport, zoning, vendor support, composting and redistribution infrastructure	livelihoods, safety, circularity	informal and small enterprises; avoid displacement	food; price dispersion; loss rates; vendor compliance
National	Social protection; FBDGs; labelling; marketing rules; reformulation; research; competition and farm policy	Affordability, NCD prevention, resilience, fair markets	Progressive financing; rural and gender analysis; transition support	Healthy-diet affordability; sodium/trans-fat exposure; farm income; concentration
International	Trade coordination; standards; climate finance; outbreak information; technology and data cooperation	Stability, safety, shared learning, climate mitigation	Policy space for low-income countries; fair benefit sharing; debt and capacity constraints	Price volatility; cross-border outbreak response; finance flows; emissions

11. Research Priorities

The next generation of food research should move beyond isolated products and short-term endpoints. Priority questions include:

- How can dietary-pattern trials measure health, affordability, acceptability, and environmental effects together over sufficient time?
- Which characteristics of ultra-processed products—energy density, eating rate, texture, additives, food matrix, palatability, marketing, or combinations—drive specific outcomes?
- How can food-safety surveillance better attribute disease to foods and hazards in low-resource settings without creating punitive barriers for informal markets?
- Which combinations of social protection, price policy, procurement, and food-environment regulation most effectively improve healthy-diet affordability?
- How do local water scarcity, biodiversity, soil carbon, nutrient pollution, and livelihoods change comparative assessments of foods beyond global average carbon footprints?
- What are the real-world environmental and nutritional effects of precision fermentation, cultivated foods, and AI-enabled production at commercial scale?

- How can indigenous and traditional food knowledge be protected from appropriation while being supported through safety, documentation, and intergenerational transmission?
- Which metrics reliably detect whether food-system transformation is equitable, culturally legitimate, and resilient rather than merely efficient?

Research infrastructure also matters. Open, interoperable data are needed on food composition, prices, sales, diets, environmental footprints, contaminants, losses, labour, and firm ownership. These datasets should be documented well enough to reveal uncertainty and bias. Funding should support long-term public-interest research, replication, implementation science, and capacity in regions where the burden is high but measurement is sparse.

12. Limitations

This review covers an exceptionally broad topic and necessarily sacrifices depth in specialized areas. It is an integrative rather than systematic review; therefore, study selection was purposive, no formal risk-of-bias instrument was applied across all disciplines, and the reference set should not be interpreted as exhaustive. Global estimates rely on models and uneven national data, and they refer to different years and denominators. Environmental averages may not represent local production systems. Nutrition evidence is vulnerable to measurement error, confounding, and substitution effects, while food-technology projections often depend on unproven future scale and energy assumptions. Cultural analysis is also limited by the impossibility of representing the diversity of food practices in a global synthesis.

The manuscript provides a publication-ready general structure, but it has not undergone external peer review. Before submission, the author should select a target journal, update the literature search, verify every reference against the final source, adapt the paper to journal instructions, and consider review by specialists in nutrition, food safety, food science, anthropology, economics, and environmental assessment.

13. Conclusion

Food sustains life, but food systems distribute life chances unevenly. They can nourish, connect, employ, and delight; they can also expose people to hunger, disease, exploitation, ecological degradation, and loss of cultural autonomy. These outcomes are not properties of “food” in the abstract. They emerge from how foods are produced, formulated, priced, promoted, regulated, prepared, shared, and discarded.

The central scientific and policy error is fragmentation. Nutrition without affordability becomes advice for those who already have options. Food safety without livelihood

support can exclude small producers and vendors. Sustainability without nutrition can reduce environmental impact while worsening deficiency. Innovation without governance can concentrate power or optimize the wrong outcomes. Culture without safety can protect harmful practices, while standardization without culture can destroy trust and diversity.

A credible food future therefore rests on an integrated definition of quality. Good food is adequate and balanced, but also safe. It is affordable and accessible, but also desirable and culturally legitimate. It is produced through dignified livelihoods, but also within ecological limits. It uses technology, but subjects novelty to public-value tests. It minimizes loss and waste, yet recognizes that efficiency alone cannot substitute for justice. Food-system transformation will succeed when these conditions are pursued together rather than traded away one at a time.

Declarations

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Consent for publication: Not applicable.

Availability of data and materials: No new dataset was generated. The manuscript relies on publicly available reports and peer-reviewed publications listed in the references.

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